The ten minimum universal sets of MacMahon cubes

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31 October 2026

Abstract

We classify universal collections of twelve six-coloured MacMahon cubes, without requiring internal face matching. A row-sum bound in Conway's tableau forces a symmetric matrix with every row and column sum equal to two. Corner coverage then forces two disjoint triangles. This proves the ten-set uniqueness conjecture stated by Johnson and Roldán, and also excludes additional minimum collections containing repeated cube varieties. The written proof is elementary and does not require exhaustive search. Exhaustive computation checks the classification and supplies additional target-count distributions. The minimum-size theorem, the complete classification and their discrete geometric interpretation are formalised in Lean 4.

Status and transparency. This research manuscript and its Lean formalisation were developed with AI assistance under the author's direction. The formal proofs have been checked by Lean; independent human expert review and publication priority beyond the cited sources remain outstanding. Details of the proof interface and AI involvement appear below.

1 Conventions and tableau facts

The six face colours, labelled $1, \ldots, 6$, remain fixed throughout the classification. Proper rotations identify cube varieties; reflections do not. There are thirty such varieties whose six faces have six different colours. In particular, the count of ten minimum collections is with fixed colour labels, not up to colour permutation; the ten collections form a single colour-permutation orbit, as checked below. A collection is *universal* if eight of its physical cubes can build a $2 \times 2 \times 2$ model of every target variety. Interior face colours need not match. Repeated varieties are allowed in the theorem; the conjecture in [1] concerns sets of distinct varieties.

Haraguchi proved that twelve is the minimum size, allowing repeated varieties, and exhibited one universal collection of that size [2, Lemma 2 and Theorem 4]. Johnson and Roldán exhibited ten such sets and conjectured their completeness [1, Theorem 3 and Conjectures 4 and 8]. Our contribution is an elementary completeness proof, including the exclusion of repeats at the minimum, and its Lean formalisation.

Following the exposition of Conway's tableau in [3, Section 3], label its entries by ordered pairs ij of distinct elements of $L = \{A, B, C, D, E, F\}$; ij corresponds to the paper's upper/lower-case name such as Ab. These tableau labels are distinct from the face-colour labels $1, \ldots, 6$. Each cube has eight oriented corner types. A corner type is an ordered triple of face colours modulo cyclic permutation, with a fixed handedness. There are forty such types.

Lemma 1 (Corner–directed-triangles correspondence)**.** *The forty oriented corner types are in bijection with partitions of L into two triples, each equipped with a directed 3-cycle. The six cube varieties containing a given corner are exactly the six arcs of its two cycles. Equivalently, cube ij contains a corner precisely when its cycles contain the arc $i \to j$.*

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Proof. Table 1 records the correspondence explicitly. For a triple written XYZ in increasing label order, write $+$ for the cycle $X \to Y \to Z \to X$ and $-$ for its reverse. The first and second signs refer to the left and right triples respectively.

To verify the table, for each corner list the cube names containing it in the standard tableau [1, Table 3], and read each name as an arc. For example, corner 256 occurs in exactly

$$Ab,\ Bc,\ Ca,\ De,\ Ef,\ Fd.$$

These give $A \to B \to C \to A$ and $D \to E \to F \to D$, so 256 occupies the $(+,+)$ column of the $ABC \mid DEF$ row. For every entry the same check gives precisely the six arcs specified by its row and column. The forty displayed triples are distinct modulo cyclic permutation and exhaust the $2\binom{6}{3} = 40$ oriented corner types. The ten rows and four columns exhaust all partitions and direction choices, proving the bijection and both directions of the incidence rule. As a further check, an arc $i \to j$ lies in exactly eight such cycle pairs: there are four choices of the third label in its cycle and two directions for the complementary cycle, recovering the eight corners of cube ij. □

Partition	$(+,+)$	$(+,-)$	$(-,+)$	$(-,-)$
$ABC \mid DEF$	256	143	134	265
$ABD \mid CEF$	465	132	123	456
$ABE \mid CDF$	345	126	162	354
$ABF \mid CDE$	235	164	146	253
$ACD \mid BEF$	364	125	152	346
$ACE \mid BDF$	356	142	124	365
$ACF \mid BDE$	153	246	264	135
$ADE \mid BCF$	165	243	234	156
$ADF \mid BCE$	136	254	245	163
$AEF \mid BCD$	263	145	154	236

Table 1: Corner types by partition and cycle directions. On each displayed triple XYZ, $+$ means $X \to Y \to Z \to X$ and $-$ means its reverse; the two signs specify the left and right triples. Corner numbers use the fixed face-colour labels of [1], modulo cyclic permutation.

The standard corner-sharing rule [4, Lemma 2.7], in tableau form [2, Propositions 1–2 and Property 2], follows from Lemma 1: two distinct varieties ij and uv share no corner exactly when $i = u$, $j = v$, or $(u, v) = (j, i)$. In every other case they share two corners. Indeed, two distinct arcs in the same directed 3-cycle must form a directed path; this fixes that cycle, leaving two orientations of the other one. For two disjoint arcs, the remaining two labels can be assigned to their respective triples in two ways, each determining both cycle directions. Common tails, common heads, and reversed arcs cannot coexist in directed 3-cycles. The target itself shares all eight corners.

Call a cube *usable* for a target when they share an oriented corner. Consequently each cube is usable for exactly 21 targets. Also, assigning eight distinct physical cubes to the eight target corner types is both necessary and sufficient: matching an oriented corner permits a proper rotation matching the three exposed faces. There are no internal constraints.

2 Necessity

Let a_{ij} be the number of copies of variety ij in a collection of size K, with $a_{ii} = 0$. Write $r_i = \sum_j a_{ij}$ and $c_j = \sum_i a_{ij}$.

Lemma 2 (Row bound)**.** *If every target has at least eight usable physical cubes in the collection, then*

$$r_i + c_j - 2a_{ij} + a_{ji} \le K - 8 \qquad (i \ne j), \qquad 3r_i \le 4K - 40, \quad 3c_i \le 4K - 40.$$

Proof. For target ij, the unusable cubes are those in its row apart from ij, those in its column apart from ij, and the copies of ji. These classes are disjoint and total $r_i + c_j - 2a_{ij} + a_{ji}$. At least eight cubes must remain usable, proving the first inequality. Summing over the five $j \ne i$ gives

$$5r_i + (K - c_i) - 2r_i + c_i = 3r_i + K \le 5(K - 8).$$

The column bound follows by the analogous sum. □

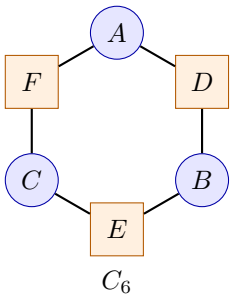
Lemma 3. *A universal collection of size twelve consists of one copy of every directed edge within each of two disjoint triples of tableau labels.*

Proof. The row bound gives $r_i \le \lfloor 8/3 \rfloor = 2$. As the six row sums total twelve, all equal two. The same holds for the columns. If $a_{ij} = 0$, the first inequality now gives $4 + a_{ji} \le 4$, so $a_{ji} = 0$. Thus support is symmetric.

Consider the undirected support graph on the six tableau labels L: an edge $\{i, j\}$ is present when $a_{ij} > 0$. Every vertex has degree one or two. If a vertex i has degree one, with neighbour j, then $a_{ij} = 2$; column i and support symmetry give $a_{ji} = 2$, so this is an isolated double-copy pair. Every degree-two component is a simple cycle, with one copy in each direction on every edge.

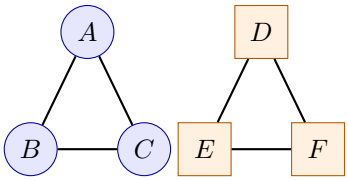
On six vertices the only possibilities are two triangles, a 6-cycle, a 4-cycle and a double-copy pair, or three double-copy pairs. All except two triangles are bipartite with parts of size three. Choose these parts as the two triples defining a corner type. Every directed edge of that corner lies within a part, whereas every selected variety joins the parts. Thus the collection contains no copy of that corner type. A target having that corner cannot be built, contradicting universality. Only two triangles remain, and their arc multiplicities are all one. □

For distinct varieties the double-copy cases cannot occur, leaving just the two support types illustrated in Figure 1.



C_6

All arcs cross $ABC \mid DEF$.
Four corner types are missing.



$C_3 \sqcup C_3$

Arcs stay within each triple.
Every target can be built.

Figure 1: The two possible degree-two supports for twelve distinct varieties. Each undirected edge represents both cube varieties ij and ji, hence six edges represent twelve cubes. Circles mark $\{A, B, C\}$ and squares mark $\{D, E, F\}$; these are tableau labels, not face colours. The left support misses all four corners for this partition (Table 1); the right support is universal. Repeated varieties allow the additional support types treated in the proof.

3 Sufficiency and classification

Lemma 4. *For any partition $L = P \sqcup Q$ into triples, the twelve directed edges within P and Q form a universal set.*

Proof. Fix target ij. Its corners are indexed by a third label $k \notin \{i, j\}$ and either direction of the cycle on the complementary triple. The first cycle is $i \to j \to k \to i$.

Suppose $i, j \in P$, and write $P = \{i, j, p\}$. For each $k \in Q$, the complementary cycle contains one arc between the two labels of $Q \setminus \{k\}$. Assign that cube to the corner. As k and the cycle direction vary, these are the six distinct directed edges within Q. The remaining two corners have $k = p$; fill them with jp and pi, both of which belong to both corner types. This gives eight distinct cubes.

Now suppose $i \in P$, $j \in Q$, with $P \setminus \{i\} = \{u, v\}$ and $Q \setminus \{j\} = \{x, y\}$. For the four corners with $k \in \{u, v\}$, the usable selected varieties are ki and one of xy, yx. Their incidence is described by a $K_{2,2}$ with vertex classes $\{ui, vi\}$ and $\{xy, yx\}$. Here the four corners are the edges, and the four candidate cube varieties are the vertices: each corner joins its two candidates. Orient the resulting 4-cycle cyclically and assign each edge to its head. Every vertex then receives exactly one edge, so each corner receives an incident candidate and all four assigned cubes are distinct. For the other four corners, the analogous four varieties are jx, jy, uv, vu. The two groups of four varieties are disjoint, completing the construction. □

Theorem 5. *The minimum size of a universal collection is twelve. Exactly ten labelled collections attain that size, even when repeated varieties are permitted. They are the two-triangle sets described above. In particular, Conjectures 4 and 8 of Johnson and Roldán [1] hold.*

Proof. Each physical cube is usable for 21 targets, so universality requires $21K \geq 30 \cdot 8 = 240$, hence $K \geq 12$. This lower bound is due to Haraguchi [2, Lemma 2]. The preceding lemmas give necessity and sufficiency at twelve. There are $\frac{1}{2}\binom{6}{3} = 10$ unordered partitions into triples. □

Corollary 6 (Twelve-cube criterion)**.** *A collection of twelve physical cubes, with repeated varieties allowed, is universal if and only if both of the following hold:*

1. *Every target has at least eight usable cubes in the collection, counting physical copies.*
2. *Every oriented corner type occurs on at least one cube in the collection.*

Proof. Universality supplies eight usable cubes for each target. Every oriented corner type belongs to some target, so it must occur in the collection.

Conversely, condition (1) gives the inequalities of Lemma 2. At size twelve these force every row and column sum to equal two and the support to be symmetric, exactly as in the proof of Lemma 3. The same degree argument leaves the four support types listed there. Each bipartite type omits all four corners associated with its balanced bipartition, contradicting condition (2). Only the two-triangle type remains, with one copy of each arc, and Lemma 4 proves it universal. □

4 Computational checks and additional results

An exhaustive enumeration of all $\binom{30}{12} = 86,493,225$ subsets returns exactly the ten sets above. The implementation also reproduces the published distribution in [1, Table 2] for all $\binom{30}{8} = 5,852,925$ eight-cube sets. Two feasibility algorithms implement Haraguchi's criteria: bipartite matching [2, Proposition 3] and the tree-component test [2, Proposition 4]. In the latter, form a compatibility multigraph whose vertices are all eight target corners, retaining isolated vertices. Each usable non-target physical cube contributes an edge joining its two compatible target

corners; parallel edges are allowed, since different cubes can have the same pair of compatible corners. This is distinct from the support graph on six tableau labels used in the necessity proof. Here cubes are edges and target corners are vertices. A connected tree component has deficit one, including an isolated vertex; a component containing a cycle has deficit zero, with two parallel edges already forming a cycle. Thus a target is buildable precisely when the number of tree components does not exceed the number of available target copies (zero or one in the subset enumerations). The table-based implementation precomputes all 2^{20} edge subsets for each target. No symmetry reduction is used in the exhaustive subset loops.

Independent geometric code generates all 24 proper cube rotations, all 30 varieties and all 40 corner types from face-colour permutations. It checks the tableau, the directed-triangle correspondence, and explicit physical placements for all 300 set–target pairs. Independently enumerating the 70 simple 2-regular graphs gives 60 six-cycles, each building 18 targets, and the ten universal triangle-pair sets. The ten sets form one colour-permutation orbit, as already observed in [1, Section 6]; our geometric check recovers this orbit. Its stabilizer has order $6!/10 = 72$.

For sets of distinct varieties the exhaustive target-count ranges are:

set size	8	9	10	11	12	13
minimum targets	0	0	1	2	8	10
maximum targets	5	9	20	22	30	30

The lower bound of one target for ten distinct varieties was proved in [3, Lemma 4.1]; the enumeration confirms that it is sharp. No set of size at most twelve builds exactly 29 targets. At size thirteen, 827,250 sets build exactly 29 targets and 53,100 are universal. These enumeration results answer the questions in [1, Section 6] about the maxima for sizes nine, ten and eleven and the possibility of building exactly 29 targets. The earlier distributions for sizes nine through twelve used random samples of 20,000 sets each; our ranges are exhaustive. Unlike the uniqueness theorem, the upper bounds here rely on the complete computational enumeration. Full histograms, example masks, source code and independent checkers accompany this draft.

5 Formal verification and reproducibility

The Lean 4 development accompanies this manuscript at https://github.com/jasondavies/macmahon-min-universal-sets. It uses Lean 4 [5] and mathlib [6], pinned to versions 4.35.0 and v4.35.0 respectively. The main results are `MacMahon.minimum_size`, `MacMahon.physical_exactly_ten`, and `MacMahon.physical_iff_two_triangles`.

An inventory is a natural-number multiplicity for each ordered pair of six tableau labels, with zero diagonal. A construction is an injective assignment of eight physical copies to the eight positions, each with a proper rotation matching its three exposed faces. The formalisation proves that the rotation table is exactly the set of determinant-positive signed-axis permutations, and that the thirty representatives and their rotations give all 720 assignments of six distinct colours to six faces, without duplication. It also proves the equivalence of corner matching and exposed-face matching.

The inequalities and symmetry argument are proved for arbitrary natural multiplicities. For the remaining classification, label permutations normalize the first row to a doubled edge or two distinct neighbours. Arithmetic proofs exclude the doubled edge and force the neighbours to close a triangle; the degree equations then determine both triangles. All finite checks use kernel reduction, rather than a trusted external computation. Explicit witnesses certify all 300 minimum-set–target constructions. The proof library contains no added axioms or unfinished proofs; guarded audits allow only Lean's standard propositional extensionality, classical choice and quotient axioms. The larger target-count enumerations above are not included in these

formal theorems. Corollary 6 is derived in the written proof; it is not separately stated in the Lean interface.

For Palomar verification, `Challenge.lean` imports only mathlib and states ten advertised results with intentional proof holes, independently of the proof library. `Solution.lean` imports the completed proofs. The comparator configuration checks equality of the statements, their permitted axioms, and kernel acceptance. Local verification outputs and reproduction commands are recorded in the repository; a successful local check is distinct from registry acceptance.

AI disclosure. OpenAI Codex assisted with the investigation, compiled solvers, geometric checks, proof argument, Lean development, manuscript and submission packaging, under the author's direction. No claim of independent human expert review is made here. The proof generators are not trusted components: their output must be accepted by Lean's kernel.

References

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